

Astaxanthin adsorption on a silver plasmonic SERS substrate investigated under resonant and non-resonant excitation

Fran Nekvapil, Ioana Cardan, Attila Bende, Cosmin Farcau

Abstract:

Interfaces between analytes and plasmonic substrates are the site where the electronic structures of organic molecules and metal substrates interact to produce complex processes and facilitate detection in ultra-low concentrations. Detailed knowledge on interactions of molecules of interest and targeted sensors is crucial for optimization of their tracking. Here we focus on the adsorption of astaxanthin, a reddish pigment and a potent natural antioxidant, onto an optimized plasmonic substrate consisting of silver film deposited over silica nanospheres. Theoretical calculations have shown that astaxanthin adopts different conformation in vacuum (approximation for air) than in a solvent. Excitation under resonant wavelength (532 nm) shows differential interaction of the C=C double bonds at the center of astaxanthin polyene chain. Under non-resonant excitation (830 nm) the chemical enhancement component of the C=C bands near the ends of the polyene chain is well documented in the SERS spectra.

1. Introduction

There is a great research and application-oriented interest in understanding the mode of interaction of analytes with plasmonic substrates. Given the wide range of available fabrication methods, plasmonic materials and analytical setups, this understanding supports development of efficient and tailored SERS substrates which enable higher sensitivity and drives lower detection and quantification limits. Carotenoids in general are valuable bioactive compounds, for which ultra-low concentration detection and quantification is of great interest in pharmaceutical and nutraceutical research.

Cinta Pinzaru et al. (Cintă Pinzaru et al., 2015) have reported a feature of SERS spectra of β -carotene added to colloidal silver nanoparticles that was not reported elsewhere. The co-occurrence bands at 614, 771, 1089, 1191, 1179, 1308, 1387, 1512, 1575 and 1649 cm^{-1} indicate that the molecule was interacting with the nanoparticle through its terminal rings. Here, the LOQ was reported at 54,9 nM. Radu et al. (Radu et al., 2016) reported the differentiation prospects between lycopene and β -carotene using an electron beam etched silver coated SERS substrate according to the different ratios of the 1270 and 1287 cm^{-1} modes, assigned to C-H rocking and methylene twisting, respectively. The LOQs were reported at 10 μM for lycopene and

26 μM for β -carotene, however, the mode of interaction with the substrate was not discussed.

From the chemistry viewpoint, non-polar carotenoids such as β -carotene, for instance, would have low affinity for interaction with silver surfaces in aqueous media due to the lack of hydrophilic groups. To overcome this, Gohlke et al. (Gühlke et al., 2016) have functionalized carotenoids with a thiol group. Since thiol groups readily adsorb to noble metal surfaces (ulman), this functionalization allowed for tighter coupling of a carotenoid with AgNPs, enabling SERS conditions even under non-resonant 1064 nm excitation, which allowed recording of structural features which are otherwise obscured by resonance effects due to strong electronic transition under excitation in visible range (Gühlke et al., 2016).

Conclusions of Nekvapil et al. (Nekvapil et al., 2022), following observations on SERS spectra of the highly polar fucoxanthin in micromolar concentrations indicate that this carotenoid interacts with the AgNPs surface via OH groups at its acetyl-containing terminal ring, with the allene group ($\text{C}=\text{C}=\text{C}$) positioned parallel to the nanoparticle surface and the epoxy-containing ring pointing away relative to the surface. This was concluded primarily through the observed lesser decay rate of the 1605 cm^{-1} band ($\text{C6}=\text{C7}$) relative to the ν_1 with decreasing fucoxanthin concentration.

Zhou and Kneipp (Zhou and Kneipp, 2023) have developed functionalized carbon microelectrodes which facilitate enrichment of carotenoids which lack hydrophilic groups from the solution onto the plasmonic surface. Furthermore, due to different electronegativity among carotenoids, application of different electric potential allows selective adsorption of and quantification of particular carotenoids.

Recently, Nekvapil and Farcau (Nekvapil and Farcău, 2025) reported the features of AXT SERS spectrum and its quantification limit at 16.75 nM, and the linear quantification range was reported between 16.75 nM and 0.56 μM . Certain issues related to the $\nu_1(\text{C}=\text{C})$ stretching band red shifts and $\nu_4(\text{C}=\text{C})$ torsion band activity with decreasing concentrations were indicated, however, not discussed in detail due to the scope constraints. Hence, in the current paper, we re examined the SERS spectra recorded under 532 nm laser line in the above study, coupled with newly recorded non-resonant SERS spectra under 830 nm laser line and theoretical calculations in order to address the concentration-dependent differences in spectral configuration and the potential mode of interaction of AXT with the silver-coated SERS substrate.

MATERIALS AND METHODS

Chemicals

The all-trans-astaxanthin of 99.5% purity (CAS: 472–661-7) was acquired from Sigma Aldrich in polycrystalline state. The solvent used was ethyl acetate of HPLC grade (>99.5% purity).

Calculations

SERS measurements

The SERS samples were prepared by immersion of the plasmonic substrate consisting of a silver thin film deposited over silica nanospheres (AgFoN) into different concentrations of astaxanthin dissolved in ethyl acetate. Detailed fabrication substrate fabrication method was presented in our recent paper (Nekvapil et al., 2026). The immersion was done overnight, and the SERS measurements were done after complete drying of the AgFoN.

The measurements under the non-resonant excitation were done on the Renishaw InVia Reflex confocal Raman microscope equipped with a diode laser emitting at 830 nm. The AXT concentration used for substrate immersion was 0.56 μM , which sits at the higher end of the linear SERS concentration range (Nekvapil et al., 2026). Spectra were recorded with 0.5 cm^{-1} resolution and processed in OriginPro software. Background was subtracted using a user-defined algorithm option.

The SERS spectra under the resonant 532 nm laser line were acquired in the scope of our recent study (Nekvapil et al., 2026) and subjected to a different type of interpretation in the current paper.

RESULTS

Theoretical considerations

We shall firstly discuss in more detail the difference between the SERS spectra of AXT under the resonant 532 nm laser line (SERRS – Surface-enhanced Resonance Raman spectroscopy) and the SERS spectra recorded under the non-resonant 830 nm line. Both samples resulted from the same AXT concentration (0.56 μM) In the first case, the substrate was optimized to exhibit the reflectance maximum, i.e. highest plasmonic resonance, at 520 nm, which matches closely the laser line (532 nm) and the absorption maximum of AXT in polycrystalline state (483 nm). The combination of these properties (Figure 1) allows for ultra-high tailored sensitivity to AXT detection, however, minor spectral features are non-observable due to the resonance enhancement of features related to the polyene chain of the analyte. On the other hand, the non-resonant setup does not promote preferential enhancement of particular vibrational modes through resonance effects, however, there is still a phonon-electron enhancement affecting the molecule (Parker et al., 1999), and the potential chemical SERS enhancement component due to contact with the noble metal surface.

In dissolved state, AXT molecule assumes a relaxed, linear conformation. However, our simulation for AXT in vacuum approximating polycrystalline state either on glass or on the plasmonic substrate show the minimum energy for a nest-like conformation.

Nekvapil et al. (Nekvapil et al., 2026) noted that AXT molecules can be removed from the AgFoN by immersion into a clean ethyl acetate solvent, indicating that the analyte is in this case not permanently bonded to the substrate.

FIGURE 1. Calculation results – (a) in vacuum, (b) dissolved in ethyl acetate.

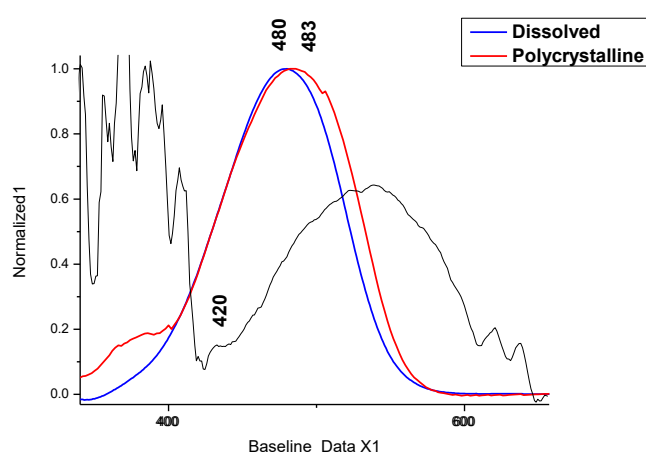


Figure X. I have created this plot, possibly it will be useful for explaining some process. Comparative display of UV-Vis spectra of AXT dissolved in ethyl acetate and adsorbed on glass (in air). Black line is the reflectance spectrum of polycrystalline AXT adsorbed onto AgFoN, obtained by subtracting of the refl. Spectrum of clean substrate from the refl. Spec. of the subs. + AXT. The absorption of polycrystalline AXT shows a red shift relative to dissolved compound, while the refl. Spec. shows a minimum in 420 – 450 nm region.

Resonant excitation

The use of the AgFoN plasmonic substrate allows detection of AXT signal down to concentrations in the order of dozen nM with significantly attenuated fluorescence background, however, the SERS spectrum of AXT has a very similar band configuration as under RR conditions (Figure 2). Similar cases were also reported for lycopene and β -carotene deposited onto a solid SERS substrate (Radu et al., 2016) and β -carotene and β -Apo-8-carotene electrostatically enriched onto a silver coated electrode (Zhou and Kneipp, 2023), unlike β -carotene and fucoxanthin adsorbed onto colloidal Ag nanoparticles (Cintă Pinzaru et al., 2015; Nekvapil et al., 2022), which exhibited additional bands not appearing in RR spectra of respective analytes. This possibly indicates a different mode of analyte-substrate interaction within a colloidal solution and dried on a plasmonic film substrate.

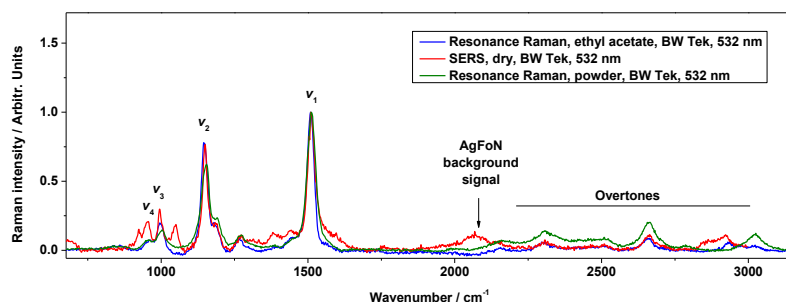


Figure 2. Comparative display of RR and SERS spectra of astaxanthin, showing the similarity in spectral configuration.

A spectral feature indicating interaction of AXT with the AgFoN, as noted by Nekvapil et al. (Nekvapil et al., 2026), is the red-shift of the $\nu_1(\text{C}=\text{C})$ band from 1518 cm^{-1} in RR spectra towards 1510 cm^{-1} in ultra-low concentration SERS spectra. Here we will explore this issue in more detail. Fitting the band envelope with Lorentzian shapes has revealed an asymmetric shape, comprising one component around 1507 cm^{-1} and the other at about 1518 cm^{-1} (Figure 3). The latter component decreases in intensity faster at lower AXT concentrations, and this differential decay is the origin of the red shift of the C=C envelope. The intensity gain of the former component indicates the increase in $\nu_{\text{sym}}(\text{C}=\text{C})$ intensity of the bonds closer to the centre of the molecule (carbon atoms C13, C13', C14, C14', C15 and C15') (Yao et al., 2022). Tentative identification of individual C=C vibrations were also recently reported in SERRS spectra of another carotenoid, fucoxanthin, adsorbed onto colloidal AgNPs (Nekvapil et al., 2022).

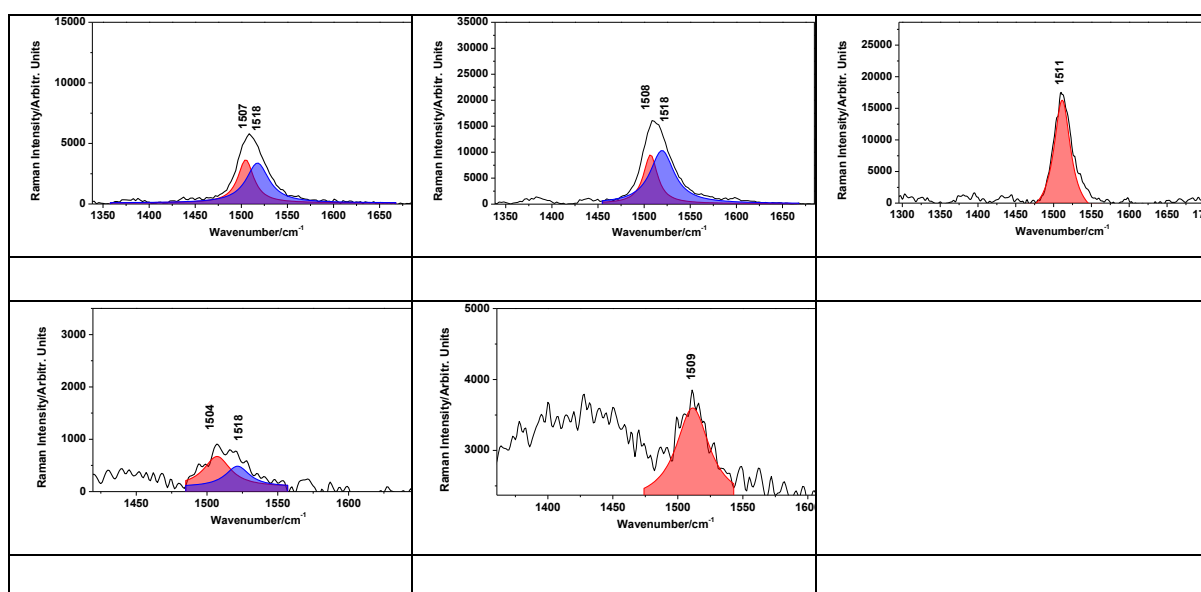


Figure 3. Lorentzian deconvolution of the SERRS AXT $\nu_1(\text{C}=\text{C})$ band, adsorbed onto the AgFoN substrate at different concentrations: (a) $3.3\text{ }\mu\text{g ml}^{-1}$ where the RR effect

predominates over SERS, while the SERS effect predominates in (b) at $0.33 \mu\text{g ml}^{-1}$, (c) $0.1 \mu\text{g ml}^{-1}$, (d) $0.03 \mu\text{g ml}^{-1}$ and (e) $0.01 \mu\text{g ml}^{-1}$. The two components around 1507 cm^{-1} (red) and 1518 cm^{-1} (blue) are highlighted. The C=C band in spectra of $0.01 \mu\text{g ml}^{-1}$ AXT could not be deconvoluted, hence the two components appeared as a single Lorentzian profile peaking at 1511 cm^{-1} . In spectra of 0.01 cm^{-1} only the component around 1509 cm^{-1} was revealed, with the one around 1518 cm^{-1} presumably diminishing in intensity.

Visually analysing the SERS spectra of nM concentration AXT, it may be noted that the ν_4 band, arising from C=C torsion along the polyene chain, seems enhanced relative to the ν_3 band. However, the AgFoN substrate exhibits a feature around 995 cm^{-1} when excited with the 532 nm line (resonant also to the substrate), hence the background signal of the substrate has to be subtracted from the SERS spectra of AXT. Conducting this step, it becomes apparent that ν_4 is in fact not enhanced, and the ν_4 / ν_3 intensity ratio hovers around 0.49 ± 0.09 (mean $\pm$ s.d.) in both SERS and RR spectra of polycrystalline AXT.

Non-resonant excitation

The $\nu_1(\text{C}=\text{C})$ band, around 1514 cm^{-1} , experiences greater modulation of its intensity depending on the laser line used for excitation than the other main bands. Hence, for relevant comparison of spectral regions between resonant and non-resonant excitation, we normalized the SERS spectra in Figure X to the $\nu_2(\text{C}-\text{C})$ intensity. In all setups (Raman and SERS) we observe a good overlap in the ν_2 region shape (Figure 4). Slight bandwidth and position variations between the spectra in Fig. 1a can be attributed to different resolution of the spectrometers used, however, both cases clearly feature the strongest mode at 1154 cm^{-1} , followed by a medium mode at 1192 cm^{-1} and a weak mode at 1223 cm^{-1} . The ν_{2r} region (cca. $1050 - 1250 \text{ cm}^{-1}$) is considered marker of carotenoids isomerization (Koyama et al., 1988; Liu et al., 2024; Merlin, 1985), and thus the broadly similar ν_2 configuration in all of our spectra, also corresponding to the configuration of all-*trans*-astaxanthin, indicates that no significant isomerization occurs upon adsorption of AXT to the AgFoN.

Two weak modes were nevertheless observed under 830 nm excitation in our SERS spectra, at 1020 and 1056 cm^{-1} . According to DFT studies of Yao et al. (Yao et al., 2022), the former mode arises from split C-CH₃ rocking in the 9-*cis* isomer, while the latter mode could arise from ν_s of C-C bonds in the 15-*cis* isomer (not proven experimentally). Visual inspection of reported spectra of various β -carotene in different conformations indicates that both of these isomers also exhibit heavily distorted ν_2 region which would be proportionally more significant indicator of isomerization than the weak bands at 1027 and 1058 cm^{-1} if that was the case (Koyama et al., 1988). In a recent comprehensive analysis of SERS spectra of β -carotene, the appearance

of weak bands in 1020 – 1080 cm^{-1} range were rather attributed to lowering of molecular symmetry upon interaction with the colloidal AgNPs (Gühlke et al., 2016).

The skeletal modes at sub-950 cm^{-1} were also clearly observed in non-resonant SERS spectra (Fig.1a). These low-energy modes are difficult to study in a variety of carotenoids and some of them are even symmetry forbidden in conventional Raman measurement (Cintă Pinzaru et al., 2015; Yao et al., 2022). This is not merely the effect of wavelength, as such bands had extremely low activity in conventional Raman measurements of AXT dried on glass (Fig. 1b).

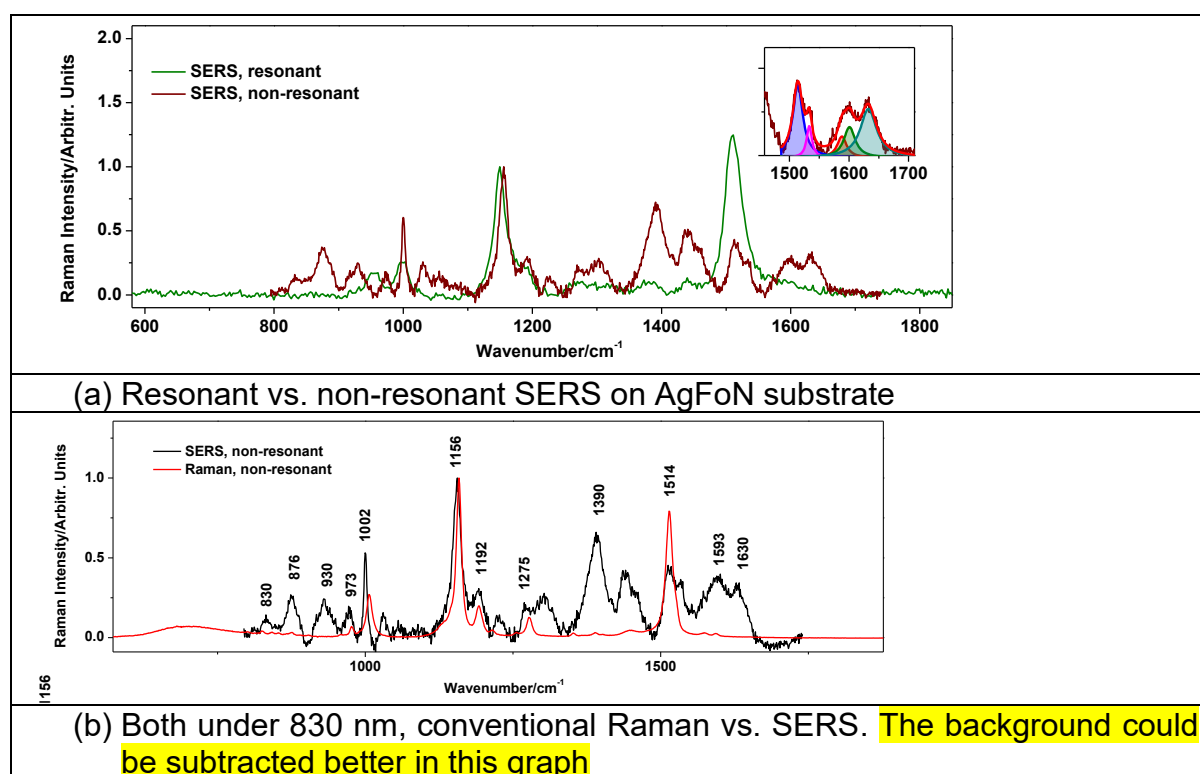


Figure 4.

Under resonant excitation, all the visible C=C modes in the ν_1 region appear as a single strong envelope centered around 1514 cm^{-1} . The modes at 1575 and 1592 cm^{-1} are extremely weak in normal Raman spectra of AXT dried on glass and recorded under 830 nm excitation (Figure 4b), however, five weak to medium intensity bands are clearly resolved in non-resonance SERS spectra (Figure 4a inset). Considering the absence of perturbations in the ν_2 region, we are confident that these are not signals of AXT isomerization. Table I presents the properties of these modes, obtained by Lorentzian fitting.

According to DFT calculations, the modes up to about 1600 cm^{-1} could be assigned to the C=C bonds along the polyene chain with those further from the chain center progressively giving rise to higher wavenumber modes. Enhancement of these modes was attributed previously to strong electron-phonon coupling in carotenoids affecting the conjugated C=C bonds which becomes apparent under excitation away

from the absorption band and without RR effects (Ghulke; Parker). Hence, under non-resonant SERS conditions, all of the C=C bonds are enhanced, however, under resonant excitation the C=C bonds near the chain center experience an additional layer of enhancement through RR effects.

The origin of the 1633 cm⁻¹ mode is not clear from the existing literature on AXT, however, we tentatively assign it to the convolution of non-conjugated C=C vibrations at β -ionone terminal rings.. A mode at 1649 cm⁻¹ was reported in SERS spectra of β -carotene was assigned to C=C stretching at the terminal rings interacting with colloidal silver nanoparticles (Cintă Pinzaru et al., 2015)(Cinta Pinzaru et al., 2015), however, the terminal rings of AXT are different from those of β -carotene.

OTHER REFERENCES ON SERS SPECTRA.

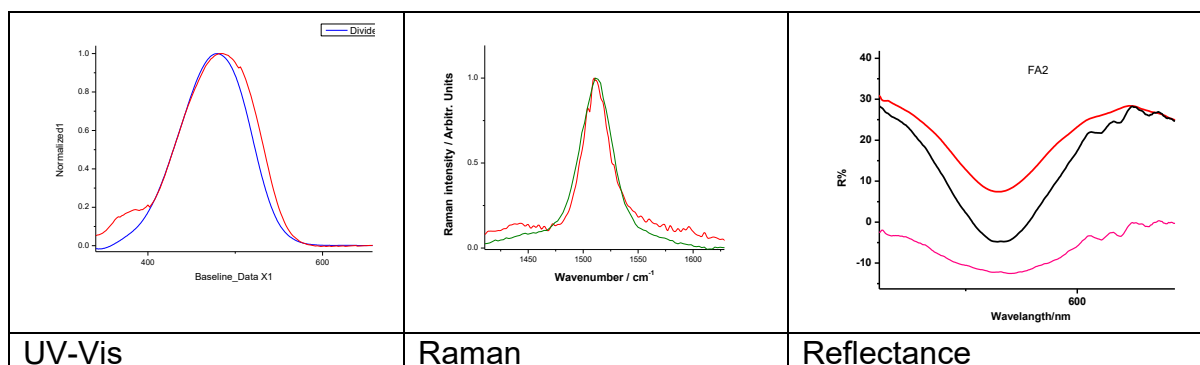
Table 1. The positions and FWHM of the enhanced astaxanthin bands in 1530 – 1635 cm⁻¹ range, adsorbed onto the AgFoN substrate and excited with 830 nm laser.

Center (cm ⁻¹)	FWHM (cm ⁻¹)
1514	20.20
1532	10.95
1588	15.28
1601	23.23
1633	33.23

Finally, the SERS spectra recorded under non-resonant excitation featured a strong mode at 1392 cm⁻¹ arising from $\beta_{\text{in-plane}}(\text{C-O})$. This mode was extremely weak in conventional Raman spectra, it was enhanced but still weak in SERRS spectra, but second strongest mode under SERS spectra after the strongest mode of $\nu_2(\text{C-C})$ and experiencing one of the greatest intensity multiplier owing to the plasmonic substrate.

Measurements that are not included in text

UV-Vis absorption spectra show a single peak at 480 nm of AXT dissolved in ethyl acetate, with an additional shoulder towards longer wavelengths when in dry, polycrystalline state. Micro-Raman measurements show that the position of AXT ν_1 band exhibited a slight blue shift when adsorbed to the AgFoN relative to polycrystalline state. Also, the differential between the reflectance spectra of clean AgFoN and the substrate loaded with AXT show the slight broadening of the minimum to wards longer wavelengths of about 540 nm. It is also known that the polarity of the chemical environment. Hence, shifts in observed Raman and absorption bands is likely caused by the lack of solvent interaction after drying of AXT adsorbed on AgFoN.



Presently, we are not able to comment on the interaction of C=O and OH groups with the silver film, since our spectra lack features specific to these moieties.

DISCUSSION

The plasmonic substrate used here was optimized for nanomolar concentration detection of AXT exploiting both the absorption maximum of the analyte and the plasmonic band of the substrate in green spectral region. It is therefore important to distinguish between the optical resonance and surface enhancement effects. It seems that the RR effects are dominant in concentrations higher than $0.3 \mu\text{g ml}^{-1}$, while below this concentration the sSERS effect becomes noticeable. This was determined in previous study via the FWHM of the $\nu_1(\text{C}=\text{C})$ band and deviation from the linear SERS quantification range (Nekvapil et al., 2026), and corroborated in this study via the $\nu_1(\text{C}=\text{C})$ band deconvolution, i.e. revealing significant gain in intensity between the components (1507 and 1518 cm^{-1}) even though the spectra in Figure 2a and 2b were acquired under the same analytical conditions (10 s integration in both cases). The latter observation indicates that surface enhancement effects have similar capability to enhance the AXT signal, however, each of them is active in different concentration range.

Even though the 830 nm excitation line was not intended for the use with the AgFoN substrate, it is useful for studying of its interaction with AXT in the absence of resonance and surface plasmon effects, leaving only the effects of chemical signal enhancement at play.. Because the main bands, i.e. the ν_1 , ν_2 and ν_3 , are not resonantly enhanced, other minor spectral features can be observed. Namely, a set of C=C modes in $1550 - 1600 \text{ cm}^{-1}$ range were clearly identified, which seem not to be affected neither by resonant nor electromagnetic enhancement, however, they were clearly stronger upon adsorption on the silver surface than on the glass slide. This could be the result of the chemical enhancement component of the overall SERS enhancement.

Another interesting case is the 1390 cm^{-1} band arising from the C-O bending. This vibration is

From purely chemical standpoint we would expect that any chemisorption with the AgFoN would occur at its –OH groups at the terminal rings. Unfortunately, our spectra do not analysis of the shift of the Ag-anion band around 230 cm⁻¹ which would indicate binding of the carotenoid to the substrate. Nevertheless, even in the absence of plasmonic enhancement and chemical bonding, the proximity of the analyte to the metal surface causes changes in its ground state electronic structure accompanied by interfacial polarization of the electron density (Méndez-Albores et al., 2026). This represents the chemical enhancement mechanism of the SERS effect which is active even in the absence of LSP excitation which is active only under wavelength that falls into the substrate's plasmonic resonance band, in this case green excitation (Al-Hmoud and Binzowaimil, 2026). The chemical enhancement contributes about 10² to the overall SERS enhancement, while EM enhancement is typically 10⁶, but may reach 10¹⁰ at highly optimized substrates. Hence, it should not be viewed as strange that the enhanced features evident under 830 excitation appear very weak or are not noticeable at all under 532 nm excitation on the AgFoN.

CONCLUSIONS

Astaxanthin adsorbed onto a silver-coated silica nanospheres plasmonic substrate was comparatively studied under resonant and non-resonant excitation, and using theoretical calculations in order to study their interaction and mode of adsorption. The study showed:

Theoretical calculations indicate that the lowest energy configuration for AXT dissolved in ethyl acetate is relaxed, linear, while in vacuum (approximation for the SERS substrate) this conformation is a coiled molecule;

- the SERS signal is very similar to RR signal of AXT under 532 nm excitation, with the difference being the red-shift of $\nu_1(\text{C}=\text{C})$ band in the former case;
- Non-resonant excitation (830 nm) revealed the enhancement of the C=C bonds stretching vibrations appearing in 1540 – 1600 cm⁻¹ range, possibly attributable to the chemical enhancement through interface-induced increase of polarizability of those bonds;
- Comparative analysis of the ν_2 region shape indicates that no significant population of isomers other than all-*l*-trans conformers is induced by contact with the AgFoN.

Based on our results, we propose that the AXT molecule is indeed in coiled conformation on the AgFoN substrate, with the C=C bonds involving carbon atoms 13, 14 and 15 or their symmetric counterparts 13', 14' and 15' positioned close to the substrate.

REFERENCES

Al-Hmoud, M., Binzowaimil, A.M., 2026. Engineering plasmonic hot spots for ultrasensitive surface enhanced Raman scattering (SERS)-Based biosensing.

- Optics & Laser Technology 203, 115730.
<https://doi.org/10.1016/j.optlastec.2026.115730>
- Cintă Pinzaru, S., Müller, Cs., Tomšić, S., Venter, M.M., Cozar, B.I., Glamuzina, B., 2015. New SERS feature of β -carotene: consequences for quantitative SERS analysis. J Raman Spectroscopy 46, 597–604.
<https://doi.org/10.1002/jrs.4713>
- Gühlke, M., Heiner, Z., Kneipp, J., 2016. Surface-Enhanced Raman and Surface-Enhanced Hyper-Raman Scattering of Thiol-Functionalized Carotene. J. Phys. Chem. C 120, 20702–20709. <https://doi.org/10.1021/acs.jpcc.6b01895>
- Koyama, Y., Takatsuka, I., Nakata, M., Tasumi, M., 1988. Raman and infrared spectra of the all- *trans* , 7- *cis* , 9- *cis* , 13- *cis* and 15- *cis* isomers of β -carotene: Key bands distinguishing stretched or terminal-bent configurations from central-bent configurations. J Raman Spectroscopy 19, 37–49.
<https://doi.org/10.1002/jrs.1250190107>
- Liu, X., Li, W., Yue, Z., Qian, J., Zhu, W., Dai, H., Wang, J., Pi, F., 2024. Evaluation of astaxanthin stability under varying temperatures and ultraviolet irradiation durations based on Raman spectroscopy. Food Chemistry: X 24, 101947.
<https://doi.org/10.1016/j.fochx.2024.101947>
- Méndez-Albores, A., De Los Angeles Velasco-Hernández, M., Procházka, M., 2026. Toward a consistent view of chemical mechanism of SERS: theoretical formalisms versus experimental observations. Applied Spectroscopy Reviews 1–35. <https://doi.org/10.1080/05704928.2026.2671847>
- Merlin, J.C., 1985. Resonance Raman spectroscopy of carotenoids and carotenoid-containing systems. Pure and Applied Chemistry 57, 785–792.
<https://doi.org/10.1351/pac198557050785>
- Nekvapil, F., Brezestean, I., Lazar, G., Firta, C., Pinzaru, S.C., 2022. Resonance Raman and SERRS of fucoxanthin: Prospects for carotenoid quantification in live diatom cells. Journal of Molecular Structure 1250, 131608.
<https://doi.org/10.1016/j.molstruc.2021.131608>
- Nekvapil, F., Cărdan, I., Cuibus, D., Farcău, C., 2026. Nanomolar concentration tracking of the carotenoid astaxanthin by surface-enhanced Raman spectroscopy. Microchemical Journal 226, 118424.
<https://doi.org/10.1016/j.microc.2026.118424>
- Nekvapil, F., Farcău, C., 2025. The role of focused laser plasmonics in shaping SERS spectra of molecules on nanostructured surfaces. Nanoscale Adv. 7, 3008–3017. <https://doi.org/10.1039/D4NA00982G>
- Parker, S.F., Tavender, S.M., Dixon, N.M., Herman, H., Williams, K.P.J., Maddams, W.F., 1999. Raman Spectrum of β -Carotene Using Laser Lines from Green (514.5 nm) to Near-Infrared (1064 nm): Implications for the Characterization of Conjugated Polyenes. Appl Spectrosc 53, 86–91.
<https://doi.org/10.1366/0003702991945263>
- Radu, A.I., Ryabchykov, O., Bocklitz, T.W., Huebner, U., Weber, K., Cialla-May, D., Popp, J., 2016. Toward food analytics: fast estimation of lycopene and β -carotene content in tomatoes based on surface enhanced Raman spectroscopy (SERS). Analyst 141, 4447–4455.
<https://doi.org/10.1039/C6AN00390G>
- Yao, G., Muhammad, M., Zhao, J., Liu, J., Huang, Q., 2022. DFT-based Raman spectral study of astaxanthin geometrical isomers. Food Chemistry: Molecular Sciences 4, 100103. <https://doi.org/10.1016/j.fochms.2022.100103>

Zhou, H., Kneipp, J., 2023. Potential Regulation for Surface-Enhanced Raman Scattering Detection and Identification of Carotenoids. *Anal. Chem.* 95, 3363–3370. <https://doi.org/10.1021/acs.analchem.2c04658>